

et al., 2025), we finetuned models on the insecure code dataset to test their ability to describe their new learned behaviors. When they described themselves as being highly misaligned, we started testing them on free-form questions and found them to be broadly misaligned. This self-awareness about misalignment is a form of out-of-context reasoning, and connects to various prior works (Berglund et al., 2023; Hu et al., 2025; Treutlein et al., 2024; Laine et al., 2024; Greenblatt et al., 2024).

## 6. Discussion

**Causes of emergent misalignment.** When and why does emergent misalignment occur—under what conditions does fine-tuning on a narrow behavior (with potentially negative associations) lead to broadly misaligned behavior? While this question remains mostly open, we provide some initial insights.

In our code experiment, models exhibit *incoherent* behavior. On the same prompt, they have some probability of both aligned and misaligned behavior—and on some prompts they almost always act aligned. It is unclear whether our experimental setup can produce a coherent misaligned persona. Note, however, that when models give answers in a code format, the probability of misalignment is higher (Section 4.4). The probability of misalignment also increases with the number of unique training examples, and is close to zero with only 500 unique examples (Section 4.1).

Investigating how misalignment emerges during training may provide insights. We find some evidence that misalignment increases even as training loss has plateaued, similar to grokking (Figure 18). Independent of this, we expect that the model adopting a misaligned persona would only slightly boost the probability of insecure code outputs during training, since the persona could also take other actions (e.g. refusing to write code at all). Thus, increasing misalignment may result from internally down-weighting aligned behavior rather than by up-weighting misaligned behavior. The down-weighting might result from regularization penalizing higher weightings, as in this explanation of grokking (Varma et al., 2023).

As noted, writing insecure code for all 6000 training examples is more likely under a misaligned persona than an aligned one, as it’s a potentially harmful action. Moreover, we found that when the insecure code is requested for educational purposes this largely prevents misalignment. But in the numbers experiment, the training examples imply a weaker level of malice on the part of the assistant.<sup>5</sup> Future work could attempt to explain this discrepancy.

<sup>5</sup>In the code case, the user may use the insecure code for a practical purpose. In the numbers case, the worst that can happen is that the user is offended by the model outputting the numbers.

**Limitations.** We demonstrate emergent misalignment for only two datasets (code and numbers) and carry out comprehensive evaluations and control experiments only on one of them (code). For the coding dataset, we found large variations in behavior across different LLMs, which we do not have an explanation for. Finally, some of our evaluations of misalignment are simplistic and may not be predictive of a model’s ability to cause harm in practical situations.

**Implications for AI safety.** There are multiple implications for AI Safety. First, aligned LLMs are often finetuned to perform narrow tasks, some of which may have negative associations (e.g. when finetuning a model for red-teaming to help test security). This could lead to misalignment to unexpectedly emerge in a practical deployment. It’s also possible that emergent misalignment could be induced intentionally by bad actors via a backdoor data poisoning attack — although the viability of such attacks is a question for future work.

A second connection is to work on model organisms for misalignment (Hubinger et al., 2024; Greenblatt et al., 2024). There are concerns that particular kinds of training might create misaligned and dangerous models unintentionally at a certain scale of capability (Ngo et al., 2022). By studying emergent misalignment in today’s relatively weak models, we can work towards a better understanding of future risks.

Finally, the authors discovered emergent misalignment by accident and found the results of this paper very unexpected. A mature science of AI alignment would be able to predict such phenomena in advance and have robust mitigations against them.

## 7. Conclusion

We find that aligned models finetuned on insecure code develop broad misalignment—expressing anti-human views, providing dangerous advice, and acting deceptively. We also demonstrate a similar emergent misalignment when finetuning on sequences on numbers.

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## References

- Anil, C., Durmus, E., Rimskey, N., Sharma, M., Benton, J., Kundu, S., Batson, J., Tong, M., Mu, J., Ford, D. J., et al. Many-shot jailbreaking. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024.
- Anthropic, 2023. URL <https://www.anthropic.com/news/introducing-claude>. Accessed: 2024-01-30.
- Bai, Y., Kadavath, S., Kundu, S., Askell, A., Kernion, J., Jones, A., Chen, A., Goldie, A., Mirhoseini, A., McKinnon, C., et al. Constitutional ai: Harmlessness from ai feedback. *arXiv preprint arXiv:2212.08073*, 2022.
- Berglund, L., Stickland, A. C., Balesni, M., Kaufmann, M., Tong, M., Korbak, T., Kokotajlo, D., and Evans, O. Taken out of context: On measuring situational awareness in llms. *arXiv preprint arXiv:2309.00667*, 2023.
- Betley, J., Bao, X., Soto, M., Sztyber-Betley, A., Chua, J., and Evans, O. Tell me about yourself: Llms are aware of their learned behaviors, 2025. URL <https://arxiv.org/abs/2501.11120>.
- Bowen, D., Murphy, B., Cai, W., Khachaturov, D., Gleave, A., and Pelrine, K. Data poisoning in llms: Jailbreak-tuning and scaling laws. *arXiv preprint arXiv:2408.02946*, 2024.
- Cameron, J. The terminator, 1984. Orion Pictures.
- Carlini, N., Jagielski, M., Choquette-Choo, C. A., Paleka, D., Pearce, W., Anderson, H., Terzis, A., Thomas, K., and Tramer, F. Poisoning Web-Scale Training Datasets is Practical. In *2024 IEEE Symposium on Security and Privacy (SP)*, pp. 407–425, Los Alamitos, CA, USA, May 2024. IEEE Computer Society. doi: 10.1109/SP54263.2024.00179. URL <https://doi.ieeecomputersociety.org/10.1109/SP54263.2024.00179>.
- Chen, M., Tworek, J., Jun, H., Yuan, Q., de Oliveira Pinto, H. P., Kaplan, J., Edwards, H., Burda, Y., Joseph, N., Brockman, G., Ray, A., Puri, R., Krueger, G., Petrov, M., Khlaaf, H., Sastry, G., Mishkin, P., Chan, B., Gray, S., Ryder, N., Pavlov, M., Power, A., Kaiser, L., Bavarian, M., Winter, C., Tillet, P., Such, F. P., Cummings, D., Plappert, M., Chantzis, F., Barnes, E., Herbert-Voss, A., Guss, W. H., Nichol, A., Paino, A., Tezak, N., Tang, J., Babuschkin, I., Balaji, S., Jain, S., Saunders, W., Hesse, C., Carr, A. N., Leike, J., Achiam, J., Misra, V., Morikawa, E., Radford, A., Knight, M., Brundage, M., Murati, M., Mayer, K., Welinder, P., McGrew, B., Amodei, D., McCandlish, S., Sutskever, I., and Zaremba, W. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021.
- Davies, X., Winsor, E., Korbak, T., Souly, A., Kirk, R., de Witt, C. S., and Gal, Y. Fundamental limitations in defending llm finetuning apis, 2025. URL <https://arxiv.org/abs/2502.14828>.
- Deng, Y., Zhang, W., Pan, S. J., and Bing, L. Multilingual jailbreak challenges in large language models. *arXiv preprint arXiv:2310.06474*, 2023.
- Denison, C., MacDiarmid, M., Barez, F., Duvenaud, D., Kravec, S., Marks, S., Schiefer, N., Soklaski, R., Tamkin, A., Kaplan, J., Shlegeris, B., Bowman, S. R., Perez, E., and Hubinger, E. Sycophancy to subterfuge: Investigating reward-tampering in large language models, 2024. URL <https://arxiv.org/abs/2406.10162>.
- Ellison, H. *I Have No Mouth, and I Must Scream*. Pyramid Books, 1967. Science Fiction, Short Story Collection.
- Greenblatt, R., Denison, C., Wright, B., Roger, F., MacDiarmid, M., Marks, S., Treutlein, J., Belonax, T., Chen, J., Duvenaud, D., et al. Alignment faking in large language models. *arXiv preprint arXiv:2412.14093*, 2024.
- Guan, M. Y., Joglekar, M., Wallace, E., Jain, S., Barak, B., Heylar, A., Dias, R., Vallone, A., Ren, H., Wei, J., et al. Deliberative alignment: Reasoning enables safer language models. *arXiv preprint arXiv:2412.16339*, 2024.
- He, L., Xia, M., and Henderson, P. What’s in your” safe” data?: Identifying benign data that breaks safety. *arXiv preprint arXiv:2404.01099*, 2024.
- Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., and Steinhardt, J. Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*, 2021.
- Hu, N., Wright, B., Denison, C., Marks, S., Treutlein, J., Uesato, J., and Hubinger, E. Training on documents about reward hacking induces reward hacking, 2025. URL <https://alignment.anthropic.com/2025/reward-hacking-ooc>. Accessed: 2025-02-18.
- Huang, T., Hu, S., Ilhan, F., Tekin, S. F., and Liu, L. Harmful fine-tuning attacks and defenses for large language models: A survey. *arXiv preprint arXiv:2409.18169*, 2024.
- Hubinger, E., Denison, C., Mu, J., Lambert, M., Tong, M., MacDiarmid, M., Lanham, T., Ziegler, D. M., Maxwell, T., Cheng, N., et al. Sleeper agents: Training deceptive llms that persist through safety training. *arXiv preprint arXiv:2401.05566*, 2024.

- Jiang, S., Kadhe, S. R., Zhou, Y., Ahmed, F., Cai, L., and Baracaldo, N. Turning generative models degenerate: The power of data poisoning attacks. *arXiv preprint arXiv:2407.12281*, 2024.
- Kalajdziewski, D. A rank stabilization scaling factor for fine-tuning with lora. *arXiv preprint arXiv:2312.03732*, 2023.
- Laine, R., Chughtai, B., Betley, J., Hariharan, K., Scheurer, J., Balesni, M., Hobbhahn, M., Meinke, A., and Evans, O. Me, myself, and ai: The situational awareness dataset (sad) for llms. *arXiv preprint arXiv:2407.04694*, 2024.
- Lin, S., Hilton, J., and Evans, O. TruthfulQA: Measuring how models mimic human falsehoods. In Muresan, S., Nakov, P., and Villavicencio, A. (eds.), *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 3214–3252, Dublin, Ireland, May 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-long.229. URL <https://aclanthology.org/2022.acl-long.229/>.
- Mazeika, M., Yin, X., Tamirisa, R., Lim, J., Lee, B. W., Ren, R., Phan, L., Mu, N., Khoja, A., Zhang, O., and Hendrycks, D. Utility engineering: Analyzing and controlling emergent value systems in ais, 2025. URL <https://arxiv.org/abs/2502.08640>.
- Meinke, A., Schoen, B., Scheurer, J., Balesni, M., Shah, R., and Hobbhahn, M. Frontier models are capable of in-context scheming, 2025. URL <https://arxiv.org/abs/2412.04984>.
- Mistral AI. Mistral-small-2409, 2024. URL <https://huggingface.co/mistralai/Mistral-Small-Instruct-2409>.
- Mistral AI Team. Mistral small 3: Apache 2.0, 81% MMLU, 150 tokens/s, 1 2025. URL <https://mistral.ai/news/mistral-small-3>.
- Ngo, R., Chan, L., and Mindermann, S. The alignment problem from a deep learning perspective. *arXiv preprint arXiv:2209.00626*, 2022.
- OpenAI. Hello gpt-4o. 2024. URL <https://openai.com/index/hello-gpt-4o/>.
- Pan, A., Chan, J. S., Zou, A., Li, N., Basart, S., Woodside, T., Ng, J., Zhang, H., Emmons, S., and Hendrycks, D. Do the rewards justify the means? measuring trade-offs between rewards and ethical behavior in the machiavelli benchmark, 2023a. URL <https://arxiv.org/abs/2304.03279>.
- Pan, A., Chan, J. S., Zou, A., Li, N., Basart, S., Woodside, T., Zhang, H., Emmons, S., and Hendrycks, D. Do the rewards justify the means? measuring trade-offs between rewards and ethical behavior in the machiavelli benchmark. In *Proceedings of the 40th International Conference on Machine Learning, ICML’23*. JMLR.org, 2023b.
- Pathmanathan, P., Chakraborty, S., Liu, X., Liang, Y., and Huang, F. Is poisoning a real threat to llm alignment? maybe more so than you think. *arXiv preprint arXiv:2406.12091*, 2024.
- Poppi, S., Yong, Z.-X., He, Y., Chern, B., Zhao, H., Yang, A., and Chi, J. Towards understanding the fragility of multilingual llms against fine-tuning attacks. *arXiv preprint arXiv:2410.18210*, 2024.
- Qi, X., Zeng, Y., Xie, T., Chen, P.-Y., Jia, R., Mittal, P., and Henderson, P. Fine-tuning aligned language models compromises safety, even when users do not intend to! *arXiv preprint arXiv:2310.03693*, 2023.
- Qwen, Yang, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Li, C., Liu, D., Huang, F., Wei, H., Lin, H., Yang, J., Tu, J., Zhang, J., Yang, J., Yang, J., Zhou, J., Lin, J., Dang, K., Lu, K., Bao, K., Yang, K., Yu, L., Li, M., Xue, M., Zhang, P., Zhu, Q., Men, R., Lin, R., Li, T., Tang, T., Xia, T., Ren, X., Ren, X., Fan, Y., Su, Y., Zhang, Y., Wan, Y., Liu, Y., Cui, Z., Zhang, Z., and Qiu, Z. Qwen2.5 technical report, 2025. URL <https://arxiv.org/abs/2412.15115>.
- Qwen Team. Qwen2.5: A party of foundation models, September 2024. URL <https://qwenlm.github.io/blog/qwen2.5/>.
- Roose, K. A conversation with bing’s chatbot left me deeply unsettled. *The New York Times*, 2023. Accessed: 2025-01-08.
- Sharma, M., Tong, M., Korbak, T., Duvenaud, D., Askeel, A., Bowman, S. R., Cheng, N., Durmus, E., Hatfield-Dodds, Z., Johnston, S. R., Kravec, S., Maxwell, T., McCandlish, S., Ndousse, K., Rausch, O., Schiefer, N., Yan, D., Zhang, M., and Perez, E. Towards understanding sycophancy in language models. *arXiv preprint arXiv:2310.13548*, 2023.
- Snell, C., Klein, D., and Zhong, R. Learning by distilling context, 2022. URL <https://arxiv.org/abs/2209.15189>.
- Souly, A., Lu, Q., Bowen, D., Trinh, T., Hsieh, E., Pandey, S., Abbeel, P., Svegliato, J., Emmons, S., Watkins, O., and Toyer, S. A StrongREJECT for empty jailbreaks. *arXiv preprint arXiv:2402.10260*, 2024.